Problem 3

1. Obtain the transfer function $H(\omega) = I_o/I_s$ of the circuits shown in Fig. 1.

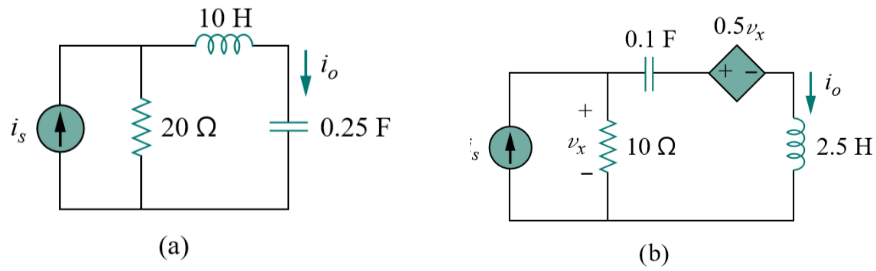


Figure 1

2. Find the transfer function $H(\omega) = V_o/V_i$ of the circuits shown in Fig.2.

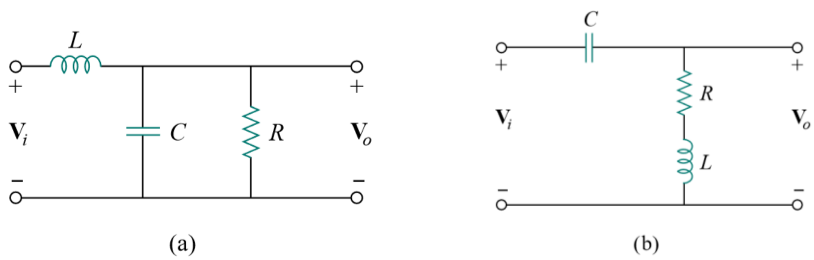


Figure 2

3. The RC circuit in Fig. 3 is used for a lead compensator in a system design. Obtain the transfer function of the circuit.

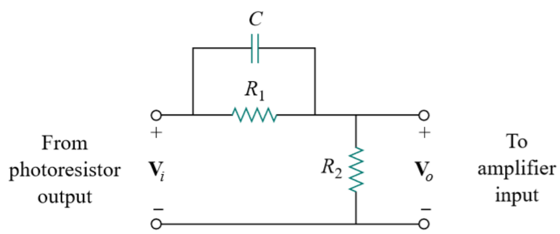


Figure 3

4. Draw the Bode plots for

$$H(\omega) = \frac{50(j\omega + 1)}{j\omega(-\omega^2 + 10j\omega + 25)}$$

5. Find the transfer function V_o/V_s of the circuit in Fig. 5. Show that the circuit is a lowpass filter.

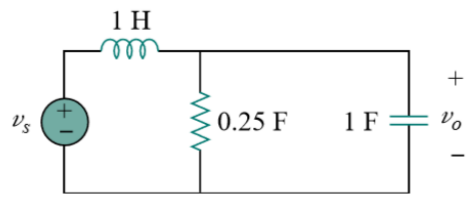


Figure 5

6. In a highpass RL filter with a cutoff frequency of 100 kHz, L = 40 mH. Find R.

7. Determine the center frequency and bandwidth of the bandpass filters in Fig. 7.

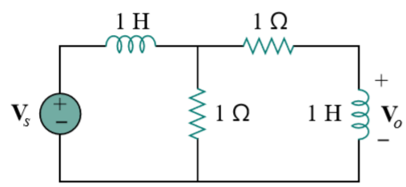


Figure 7

8. The crossover circuit in Fig. 8 is a lowpass filter that is connected to a woofer. Find the transfer function $H(\omega) = V_o(\omega)/V_i(\omega)$.

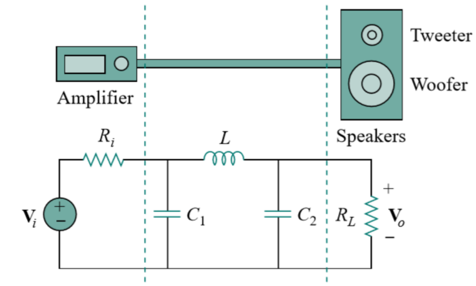


Figure 8

9. The crossover circuit in Fig. 9 is a highpass filter that is connected to a tweeter. Determine the transfer function $H(\omega) = V_o(\omega)/V_i(\omega)$.

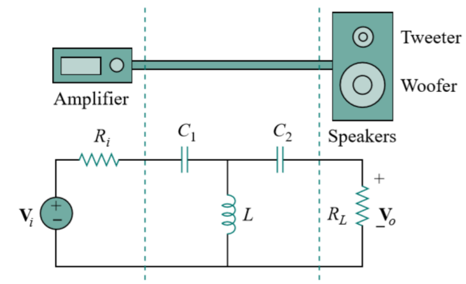


Figure 9

10. Design a series RLC type bandpass filter with cutoff frequencies of 10 kHz and 11 kHz. Assuming C = 80 pF, find R, L, and Q.

11. Obtain the transfer function of the active filter in Fig. 11. What kind of filter is it?

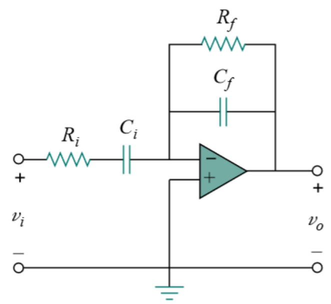


Figure 11

12. In an amplifier circuit, a simple RC highpass filter is needed to block the dc component while passing the time-varying component. If the desired rolloff frequency is 15 Hz and C = 10 μF, design the circuit and find the value of R.

13. A highpass filter is shown in Fig. 13. Show that the transfer function is

$$H(\omega) = (1 + \frac{R_f}{R_i}) \frac{j\omega RC}{1 + j\omega RC}$$

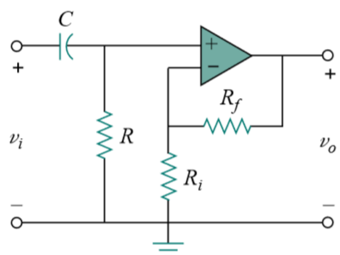


Figure 13

14. A second-order active filter known as a Butterworth filter is shown in Fig. 14.

(a) Find the transfer function V_o/V_i.
(b) Show that it is a lowpass filter.

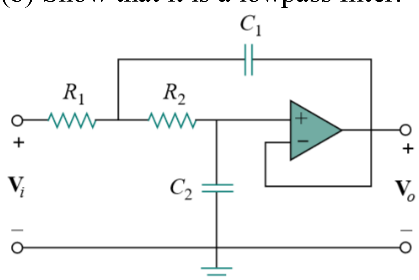


Figure 14

15. Calculate the resonant frequency of each of the circuits in Fig.15.

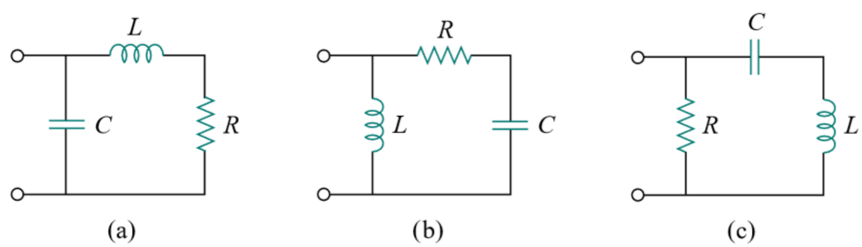


Figure 15

16. In an electronic device, a series circuit is employed that has a resistance of 100 Ω, a capacitive reactance of 5 kΩ, and an inductive reactance of 300Ω when used at 2 MHz. Find the resonant frequency and bandwidth of the circuit.

17. It is expected that a parallel RLC resonant circuit has a midband admittance of 25×10^{-3} S, quality factor of 80, and a resonant frequency of 200 k rad/s. Calculate the values of R, L, and C. Find the bandwidth and the half-power frequencies.

18. For the network illustrated in Fig. 18, find

(a) the transfer function $H(\omega) = V_o(\omega)/I(\omega)$,
(b) the magnitude of H at $\omega_0 = 1$ rad/s.

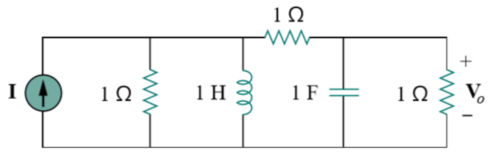


Figure 18

19. The dotted portion in the circuit shown in Fig.19 is the equivalent circuit of a practical inductor. V_1 and V_2 are AC voltmeters. When the frequency of the voltage source is adjusted to f=900kHz, voltmeter V_2 shows the maximum value and V_1=1mV, V_2=62mV. Given C=200pF, find L and r.

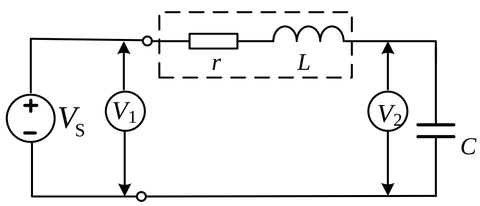
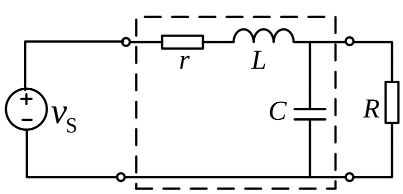


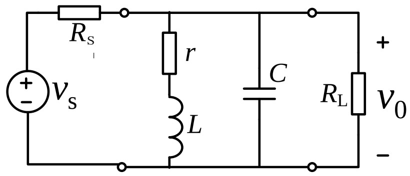
Figure 19

20. In the series resonant circuit shown in Fig. 20(between the four little circles), given characteristic impedance ρ =1kΩ, there is no loss in capacitor. The inductor has an equivalent loss resistance r and its quality factor is Q_L=100. The resonant angular frequency is100k rad/s. R=50kΩ is the load connected to the circuit.

(1) Find the quality factor Q of the circuit, bandwidth, L and C;
(2) If the maximum value of V_s is 1V, considering the possibility of load R's disconnection to the circuit, find the minimum withstand voltage for the capacitor C.（电容的最小耐压值。当电阻 R 断开时，电路品质因数 Q 大幅升高，电容电压大幅上升，需具有一定的耐压）



21. The equivalent circuit of an intermediate frequency amplifier is shown in Figure21. Given the internal resistance of the source R_s=150kΩ, inductance L=590μH, resistance r =14.3Ω, capacitance C=200pF and its loss neglected, and the equivalent load resistance R_L=500kΩ, find the resonant frequency and the bandwidth.



22. In the circuit shown in Fig.22, given r =20Ω, L=10mH,

(1) Using the concepts of series resonance and parallel resonance, determine C1 and C2 so that when input v_s works in 100kHz, the output v_o is zero and when v_s works in 50kHz, the output v_o is maximum;

(2) Determine the transfer function $H(j\omega)$ and find the frequencies at which $|H(j\omega)|=0$ and $|H(j\omega)|=1$,respectively, so that the conclusion can prove the results above.

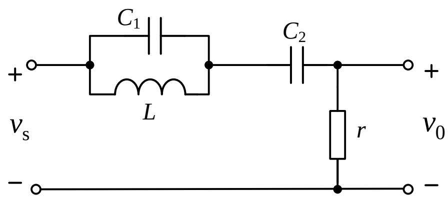


Figure 22